LIFTING A BOUNDED COHERENT OBJECT FROM AN OPEN TO A DELIGNE SYSTEM, UNIQUELY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0G4L (sources/stacks-duality/pinned-original-source.tex line 8103).

Allowed standard prerequisites (editorial).

- The colimit of Hom out of a Deligne system and the pro-category description. Same chapter, Lemma labelled lemma-lift-map, cited at proof lines 8144-8146, and Categories, Remark labelled categories-remark-pro-category-copresheaves, cited at proof lines 8146-8150. Gives part (1).
- Lifting a bounded coherent object from an open. Derived Categories of Schemes, Lemma labelled perfect-lemma-lift-coherent, cited at proof lines 8156-8158. Produces K' restricting to K.
- Representation of bounded coherent objects by complexes of coherent modules. Derived Categories of Schemes, Proposition labelled perfect-proposition-DCoh, cited at proof lines 8158-8160. Gives the complex whose ideal powers define the Deligne system.

Source reference. Original lemma 0G2H: sources/stacks-duality/pinned-original-source.tex, source lines 7920–7920. Same-chapter pointer into the author environment lemma at source line 7920; invoked at proof line 8121 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0G2X: sources/stacks-chow/tags.txt, source lines 20072–20072. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8124 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0G48: sources/stacks-chow/tags.txt, source lines 20118–20118. Cross-chapter standard prerequisite located in the project tag list

Date: stacks-project:duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

staged in the local snapshot; invoked at proof line 8129 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original proposition 0FDB: sources/stacks-chow/tags.txt, source lines 19211–19211. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 8130 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let $j : U \to X$ be an open immersion of Noetherian schemes.*

(1) *Let (K_n) and (L_n) be Deligne systems. Let K and L be the values of the constant systems $(K_n|_U)$ and $(L_n|_U)$. Given a morphism $\alpha : K \to L$ of $D(\mathcal{O}_U)$ there is a unique morphism of pro-systems $(K_n) \to (L_n)$ of $D^b_{Coh}(\mathcal{O}_X)$ whose restriction to U is α.*
(2) *Given $K \in D^b_{Coh}(\mathcal{O}_U)$ there exists a Deligne system (K_n) such that $(K_n|_U)$ is constant with value K.*
(3) *The pro-object (K_n) of $D^b_{Coh}(\mathcal{O}_X)$ of (2) is unique up to unique isomorphism (as a pro-object).*

Proof. Part (1) is an immediate consequence of Lemma 0G2H and the fact that morphisms between pro-systems are the same as morphisms between the functors they corepresent, see Categories, Remark 0G2X.

Let K be as in (2). We can choose $K' \in D^b_{Coh}(\mathcal{O}_X)$ whose restriction to U is isomorphic to K, see Derived Categories of Schemes, Lemma 0G48. By Derived Categories of Schemes, Proposition 0FDB we can represent K' by a bounded complex $\mathcal{F}^\bullet$ of coherent $\mathcal{O}_X$-modules. Choose a quasi-coherent sheaf of ideals $\mathcal{I} \subset \mathcal{O}_X$ whose vanishing locus is $X \setminus U$ (for example choose $\mathcal{I}$ to correspond to the reduced induced subscheme structure on $X \setminus U$). Then we can set K_n equal to the object represented by the complex $\mathcal{I}^n\mathcal{F}^\bullet$ as in the introduction to this section.

Part (3) is immediate from parts (1) and (2). □